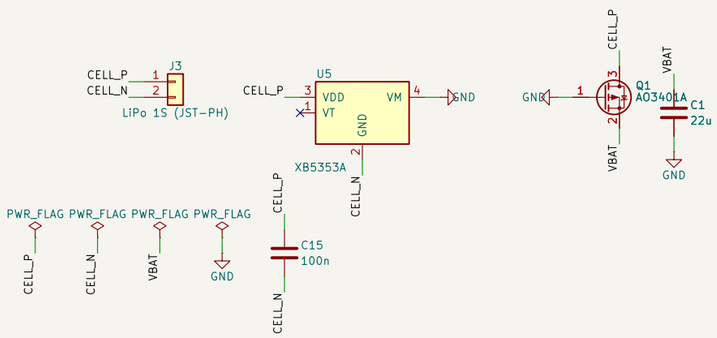
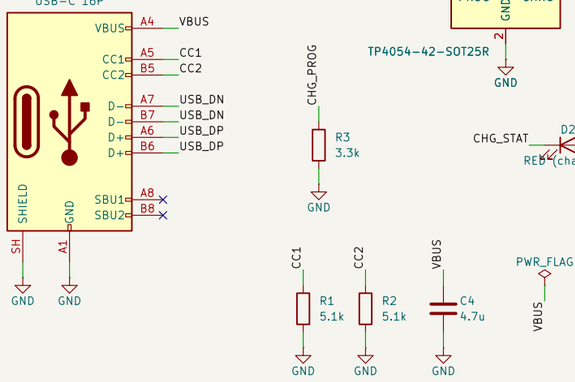
Battery, protection, reverse polarity



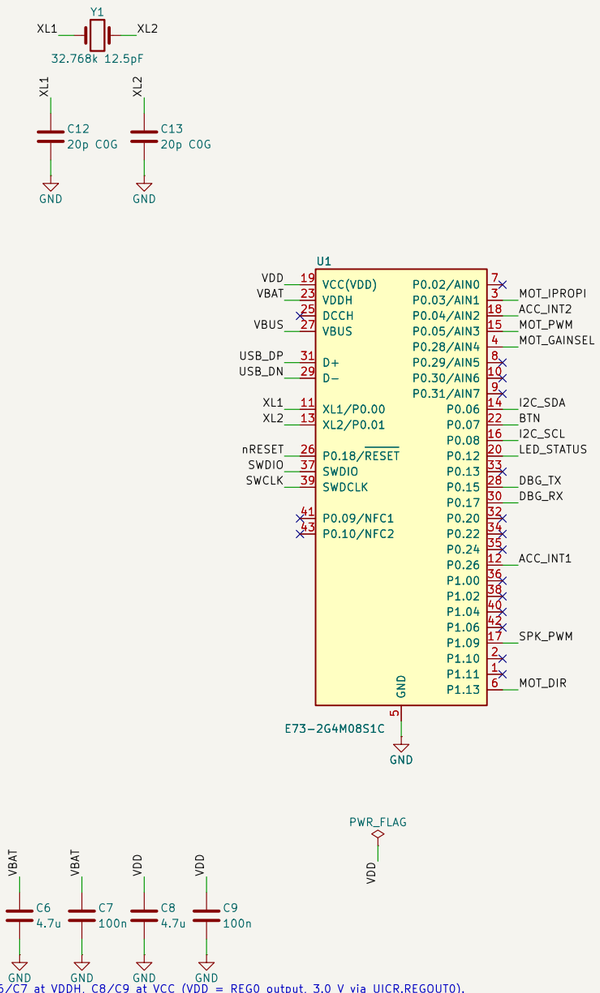
U5: VM (pins 4,5) is pack minus = board GND. Keep CELL_N -> U5 -> GND short and wide.
Q1: body diode conducts until U5 closes, then the channel turns fully on (Vgs = −Vcell).
Reversed cell: Q1 blocks the high side, U5 blocks the low side.

USB-C (sink) and LiPo charger



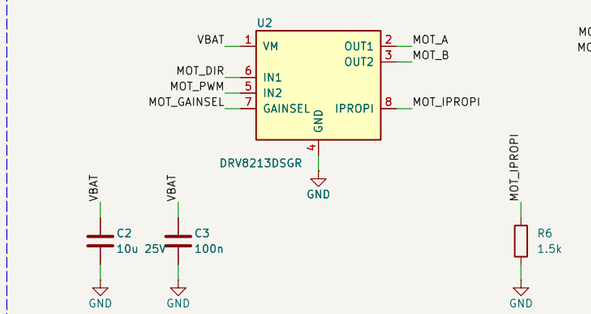
I_chg = 1000 V / R3 = ~300 mA. C4 sits between J1, U4 VCC and U1 VBUS.

nRF52840 module (Zigbee / BLE / USB)



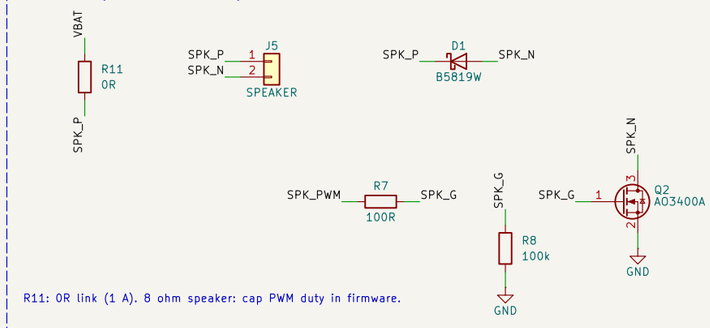
C6/C7 at VDDH, C8/C9 at VCC (VDD = REG0 output, 3.0 V via UICR.REGOUT0).
REG0 runs as LDO (DCCH unconnected). No 32 kHz crystal inside the module: Y1 is required.
PWM nets avoid the low-frequency pins near the radio (P0.02–P0.05, P0.28–P0.31, P1.1x).

Motor driver (DRV8213, current sense)



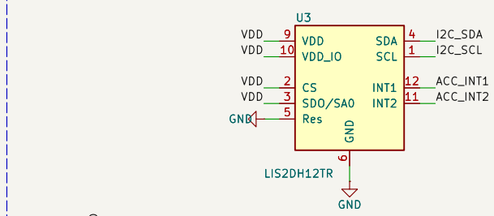
I_limit = 0.51 V / (R6 x A_IPROPI): GAINSEL low 1.66 A, Hi-Z 324 mA, high 69 mA.
Auto-sleep (20 nA) when IN1 = IN2 = 0. Keep the 100 nF cap on the motor tabs.

Speaker (optional, PWM)



R11: 0R link (1 A). 8 ohm speaker: cap PWM duty in firmware.

Accelerometer (wake-on-motion)



CS = VDD (I2C). SA0 = VDD -> address 0x19. I2C uses the nRF internal pull-ups (~13k, 100 kHz).

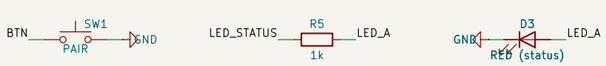
Notes

One board for all three toys (bird / fish / ball). 1S LiPo, protected or not.

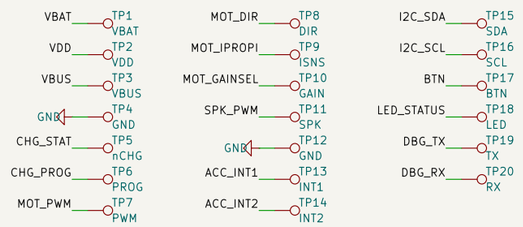
Battery voltage: SAADC VDDHDIV5 (internal), no divider.
Charging detect: nRF USBDETECTED (VBUS), CHRG drives D2 only.
Motor current: SAADC AIN1 (P0.03) on MOT_IPROPI, 0.6 V ref, gain 1.
GAINSEL: drive low / drive high / input-disconnected = Hi-Z.
Motor: DIR (IN1, static) + PWM (IN2): fwd DIR=0 PWM=duty, rev DIR=1 PWM=1−duty.
Speaker: PWM sample playback on SPK_PWM (~62.5 kHz carrier).

Hand-soldered, not JLC-assembled: J3 battery, J4 motor, J5 speaker.
J2 SWD: 1.27 mm pad row, unpopulated (first flash only, then USB DFU).
H1, H2: M2 mounting holes (NPTH), diagonal corners; ISO 7380 button head (3.5 mm).

Button, status LED, SWD



Test points (back, 1 mm pads, not assembled)



Value = back-silk label. DBG_TX/RX: spare GPIOs, 3 V UART log port (P0.15 / P0.17).
PROG: I_chg = V x 300 mA. ISNS: I_motor = V / (A_IPROPI x 1.5k).

H1 M2 H2 M2

One board for bird / fish / ball — see docs/hardware-design.md
nRF52840 (E73) + DRV8213 + LIS2DH12 + TP4054 + XB5353A, 1S LiPo
zigbee_toy

Sheet: /		
File: zigbee_toy.kicad_sch		
Title: Zigbee cat toy controller		
Size: A3	Date: 2026-10-19	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/1